Uppsala University



Nuclear Power Technical Systems

1FA410

Math simulation of a nuclear reactor

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Contents

1 Introduction **2**

2 Method **2**
2.1 Assumptions 2
2.2 The five factor formula 3
2.2.1 Thermal fission factor (η) 4
2.2.2 Fast fission factor (ϵ) 4
2.2.3 Non leakage factor (P) 4
2.2.4 Resonance escape probability (p) 5
2.2.5 Thermal Utilization Factor (f) 5
2.3 Fuel usage 7

3 Results **9**
3.1 Results of five factor formula 9
3.2 Safety Analysis and Void Coefficient 10

4 Discussion **12**

5 Conclusion **14**

6 Appendix **14**
6.1 Python-code 14

1 Introduction

The power output of a nuclear reactor is determined by the fission rate within its fuel, which relies on a continuous chain reaction. To maintain a steady power level, each fission event must yield exactly one neutron that successfully splits another uranium nucleus. In reality, not all released neutrons contribute to new fissions. For example, neutrons can be lost when non-fissile isotopes (reactor poison) capture them. To evaluate this balance across neutron generations, the effective multiplication factor (k) is used. When $k = 1$, the reactor's power output is constant, and the system is referred to as "critical". The objective of this assignment is to develop a preliminary reactor design that achieves criticality at a full-power equilibrium and can sustain operation until the next scheduled exchange of fuel.

2 Method

2.1 Assumptions

The following assumptions have been made:

Parameter	**Value**
Reactor Type	BWR
Thermal Power	2540 MW
Core Dimensions (H × D)	3.7 m × 5.95 m
Number of Fuel Assemblies	648
Operating Cycle	12 months

Table 1: Assumptions made for reactor parameters (taken from Ringhals 1)

Parameter	**Value**
Fuel Material	UO_2
Enrichment (^{235}U)	3.0 %
Fuel Density (ρ_{UO_2})	10.97 g/cm^3
Control Rod Material	B_4C
Control Rod Density (ρ_{B_4C})	2.52 g/cm^3
Number of Control Rods	157
Calibration Factor	0.15

Table 2: Assumptions made for fuel- and control rod parameters

Parameter	**Value**
Moderator Material	H_2O (light water)
Temperature	286 °C
Void Fraction	40 %
Volume Ratio (V_m/V_u)	3

Table 3: Assumptions for moderator parameters

Parameter	**Value**
Energy per Fission	$3.2 \cdot 10^{-11}$ J
Effective Resonance Integral ($\sigma_{I,abs}$)	20 barn
Slowing-down Power ($\xi\Sigma_s$)	0.8 cm^{-1}
Fission Cross Section ($\sigma_{f,235}$)	596 barn
Absorption H_2O (σ_{a,H_2O})	0.664 barn
Cumulative Iodine Yield (γ_I)	6.437 %
Independent Xenon Yield (γ_{Xe})	0.1016 %

Table 4: Physical cross sections and constants

2.2 The five factor formula

The effective multiplication factor (k_{eff}) for the reactor core is calculated using the five-factor formula:

$$k_{eff} = \eta \cdot \epsilon \cdot p \cdot f \cdot P \tag{1}$$

The individual parameters are mathematically defined and implemented as follows:

2.2.1 Thermal fission factor (η)

The calculation of the thermal fission factor in this assignment (η) is based on the fundamental assumption that Uranium-235 remains the sole fissile isotope throughout the reactors entire cycle. The model explicitly neglects the transmutation of Uranium-238 into fissile Plutonium-239, a process that naturally occurs during the fuel cycle and would otherwise contribute significantly to thermal fissions at the end of the cycle. The average number of neutrons released per thermal fission ($\bar{\nu}$) is treated as a constant value of 2.43, representing pure U-235.
Assuming Uranium-235 is the sole fissile isotope and thermal neutron capture occurs only in U-235 and U-238, η is calculated as:

$$\eta = \bar{\nu} \frac{e\sigma_{f,235}}{e\sigma_{a,235} + (1-e)\sigma_{a,238}} \tag{2}$$

where e is the enrichment fraction of ^{235}U, $\bar{\nu}$ is the average number of neutrons produced per thermal fission, and σ_a represents the microscopic absorption cross sections (fission + capture).

2.2.2 Fast fission factor (ϵ)

The fast fission factor (ϵ) accounts for the additional neutrons produced by fissions in ^{238}U caused by high-energy neutrons before they are thermalized. In this design, ϵ is assumed to be a constant value of 1.06, which has been assumed continuously throughout the course for BWR reactors and was also deemed an acceptable approximation in the instructions for this assignment.

2.2.3 Non leakage factor (P)

The total non-leakage probability (P) represents the fraction of neutrons that do not escape the physical boundaries of the reactor core during both the slowing-down and thermal diffusion phases. P is assigned a constant assumed value of 0.97 which has been assumed continuously throughout the course.

2.2.4 Resonance escape probability (p)

The calculation of the resonance escape probability (p) relies on a simplified exponential model utilizing a constant effective resonance integral ($\sigma_{I,abs}$) of 20 barn. The macroscopic slowing-down power of the moderator ($\xi\Sigma_s$) is assumed to be a constant $0.8\,\mathrm{cm}^{-1}$. This static value implicitly assumes an epithermal microscopic scattering cross section of approximately 44 barn for the water molecule (which has been assumed in the course continuously), and it does not dynamically account for the significant local variations in moderator density caused by the 40 % void fraction within the boiling water reactor core.

The probability that a neutron escapes resonance capture during moderation is calculated using the effective resonance integral ($\sigma_{I,abs}$) and the slowing-down power ($\xi\Sigma_s$):

$$p = \exp\left[-(1-e)\left(\frac{N_U\sigma_{I,abs}}{\xi\Sigma_s}\right)\left(\frac{V_m}{V_f}\right)^{-1}\right] \tag{3}$$

where N_U is the total atomic number density of uranium, and V_m/V_f is the volumetric ratio of the moderator to the fuel.

2.2.5 Thermal Utilization Factor (f)

The calculation of the thermal utilization factor (f) uses a simplified model where cross sections are scaled by their volume fractions. A primary assumption is that thermal neutrons are only absorbed by the fuel (U-235 and U-238), the moderator (assuming zero absorption for oxygen), Xenon-135 as the only reactor poison, and the B_4C control rods. This means absorption from Zircaloy cladding, internal structures, and other fission poisons like Samarium-149 is ignored. Furthermore, since a simple volume ratio cannot capture spatial effects like self-shielding around the control rods, an empirical scaling factor (0.15) is applied to the control rod volume. This factor is used simply to achieve a realistic value for the thermal utilization factor.

The thermal utilization factor is derived from the macroscopic absorption cross sections (Σ_a) weighted by their relative volume fractions ($V_{rel,i} =$

V_i/V_{fuel}):

$$f = \frac{\Sigma_{a,fuel}}{\Sigma_{a,fuel} + \Sigma_{a,H_2O} V_{rel,mod} + \Sigma_{a,Xe} V_{rel,Xe} + \Sigma_{a,B_4C} V_{rel,rods}} \tag{4}$$

To evaluate this utilization factor, the macroscopic cross-sections of the dynamic components must first be defined, beginning with the Xenon-135 poison. The stable equilibrium concentration of Xenon-135 is derived by dividing its production rate by its loss rate. The numerator represents the production, where Xenon-135 is formed both directly from fission and indirectly through the decay of Iodine-135, combined and multiplied by the reactor's total fission rate. The denominator accounts for how Xenon-135 is simultaneously removed from the core, partly through natural radioactive decay and partly through neutron absorption, or "burnup." During continuous operation, these two main processes balance each other out, and the resulting quotient provides the steady-state density established in the reactor:

$$N_{Xe} = \frac{(\gamma_I + \gamma_{Xe})\Sigma_{f,fuel}\phi}{\lambda_{Xe} + \sigma_{a,Xe}\phi} \tag{5}$$

Beyond reactor poisons, the thermal utilization factor is highly dependent on the physical ratio of the competing absorbing materials. Because the control rods are dynamically inserted during operation to maintain criticality, they physically displace the surrounding water. Consequently, the volumes of both the control rods and the available moderator must be continuously recalculated as a function of the insertion depth.

The total physical volume of the inserted control rods is calculated based on the standard cruciform (cross-shaped) design of BWR control blades. This shape essentially consists of two intersecting rectangular plates, which is represented by the factor of 2 in the total rod volume equation:

$$V_{rods} = N_{rods} \cdot (2 \cdot L_{rod} \cdot w_{blade} \cdot t_{blade}) \tag{6}$$

where L_{rod} is the actual inserted length of the rods based on the insertion percentage.

Because Boron Carbide (B_4C) is an exceptionally strong neutron absorber, the thermal neutron flux drops precipitously at the surface of the blade. Most neutrons are absorbed in the outermost layers, meaning the

inner volume of the control rod sees very few neutrons. Using the raw physical volume would severely overestimate the rod's negative reactivity worth. To account for this spatial self-shielding (flux depression), a scaling factor ($S_{shielding}$) is applied to determine the effective relative control rod volume compared to the fuel:

$$V_{rel,rods} = \frac{V_{rods}}{V_{fuel}} \cdot S_{shielding} \tag{7}$$

Once the volume of the solid fuel and the inserted control rods are known, the remaining empty space within the core's total cylindrical boundary (V_{core}) is assumed to be filled entirely by the water moderator. The total available moderator volume is simply the remainder:

$$V_{mod} = V_{core} - V_{fuel} - V_{rods} \tag{8}$$

Finally, this value is normalized against the fixed fuel volume to establish the relative moderator ratio. This ratio determines how much water is available to slow down and absorb neutrons relative to the uranium, which directly dictates the thermal utilization factor (f):

$$V_{rel,mod} = \frac{V_{mod}}{V_{fuel}} \tag{9}$$

2.3 Fuel usage

To accurately model the reactor's neutronic behavior over its operational lifetime, the depletion of the fissile inventory (burnup) must be accounted for. The total thermal energy extracted during the 1.5-year operating cycle is calculated assuming continuous steady-state operation at the full thermal power of 2540 MW. By dividing this total energy yield by the average recoverable energy per fission ($3.2 \cdot 10^{-11}$ J), the absolute number of fissions required per cycle is determined.

The methodology relies on the simplified assumption that all thermal power is generated exclusively by the fission of Uranium-235. This approach conservatively neglects the minor energy contributions from Uranium-238 fast fissions, as well as the breeding and subsequent fission of Plutonium-239.

The initial inventory of ^{235}U nuclei is derived from the total core UO_2 mass

(roughly 125 tons) and the starting enrichment of 3.0 %. To establish the End of Cycle (EOC) state, the total number of fissions is subtracted directly from this initial ^{235}U inventory. The remaining number of ^{235}U nuclei is then divided by the total number of uranium nuclei to calculate the EOC enrichment.

3 Results

3.1 Results of five factor formula

We have modelled a BWR-type reactor using the five-factor formula to evaluate its neutronic behavior over a theoretical 1.5-year operating cycle at a continuous thermal power of 2540 MW. Our results demonstrate that a stable, critical reactor design ($k_{eff} = 1$) is achieved at the chosen equilibrium void fraction of 40 %, both at the Start of Cycle (BOC) and End of Cycle (EOC).

Parameter	**BOC (Start)**	**EOC (End)**
k_{eff}	1.00000	1.00000
η (Thermal fission)	1.84851	1.68619
ϵ (Fast fission)	1.06000	1.06000
p (Resonance escape)	0.82055	0.81830
f (Thermal utilization)	0.64120	0.70486
P (Non-leakage)	0.97000	0.97000

Table 5: Calculated values for the five-factor formula at the start and end of the cycle.

Table 5 displays the explicit values for each of the parameters in the five factor formula. At BOC, the fresh UO_2 fuel has an assumed initial enrichment of 3.0 % ^{235}U. This relatively high fissile concentration provides significant excess reactivity, primarily yielding a high thermal fission factor (η). To compensate for this and prevent a supercritical state, the B_4C control rods must be inserted to a depth of 19.9 %. The rods act as a strong thermal neutron sink, deliberately reducing the thermal utilization factor (f) to suppress the excess neutrons and maintain $k_{eff} = 1$.

Throughout the 1.5-year operating cycle, continuous energy extraction results in some fuel depletion. The fissile ^{235}U is consumed, which lowers the core's enrichment to 1.65 % at EOC. Consequently, the macroscopic fission cross section of the fuel decreases, which natively lowers both η and the resonance escape probability (p). Furthermore, the steady-state operation

builds up an equilibrium concentration of Xenon-135, introducing a constant parasitic absorption penalty into the thermal utilization factor.

To counteract the loss of reactivity from fuel burnup and Xenon poisoning, the control rods must be gradually withdrawn to maintain a critical chain reaction. Our model calculates that at EOC, the rods must be retracted to a 5.3 % insertion depth. By doing that f increases just enough to balance the degraded fuel quality, successfully maintaining $k_{eff} = 1$ at the end of the operating cycle aswell.

3.2 Safety Analysis and Void Coefficient

For a boiling water reactor (BWR) to maintain inherent safety and stability, it must operate with a negative void coefficient. A negative void coefficient directly correlates to a negative moderator temperature coefficient. Graphically, this requirement dictates that the effective multiplication factor (k_{eff}) curve must exhibit a distinctly negative slope at the operational void fraction. Under such conditions, an unintended increase in core temperature and subsequent steam production (voiding) introduces negative reactivity into the system, naturally suppressing the nuclear chain reaction and stabilizing the power output.

At the Start of Cycle (BOC), the current reactor design exhibits excellent inherent safety. As illustrated by the solid blue curve in 1, k_{eff} peaks at a low void fraction and demonstrates a steep negative slope at the operational design point of 40 % void. If a thermal transient were to increase the void fraction from 40 % to 50 %, the effective multiplication factor drops significantly. This behavior confirms that the core is safely under-moderated at BOC, providing a strong, self-regulating safety mechanism.

At the End of Cycle (EOC), represented by the dashed red curve in 1, the void coefficient profile shifts significantly due to fuel depletion and changing core properties. The curve displays a positive slope from a fully flooded state (0 % void) up to approximately 35 % void. While this indicates a positive void coefficient regime at lower void fractions—which would require careful reactivity management during reactor start-up or low-power operation—the reactor safely transitions out of this regime before reaching nominal power.

Crucially, at the chosen operational target of 40 % void, the EOC curve has already passed its local maximum. Although the negative slope at this specific point is less pronounced compared to the BOC state, it confirms that the reactivity feedback remains negative at the equilibrium operating point. Consequently, any transient temperature increase that generates additional void beyond 40 % will successfully reduce k_{eff} and inherently suppress the power output.

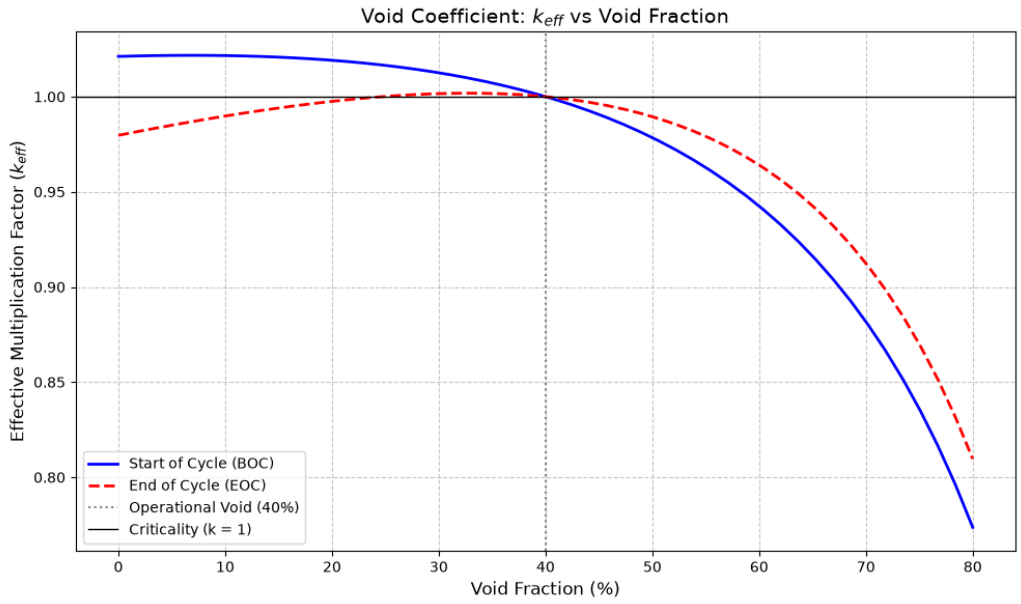


Figure 1: Graph showcasing the void in the designed BWR

4 Discussion

The preliminary design and neutronic analysis of the boiling water reactor (BWR) relied on optimizing specific core parameters to maintain a critical state ($k_{eff} = 1$) throughout the fuel cycle. Because the core was modeled using mainly the five-factor formula, several physical phenomena required parameterization and numerical optimization to yield realistic results.

The primary optimization in the code was the dynamic adjustment of the control rod insertion depth. To continuously balance the excess reactivity provided by the fresh 3.0 % ^{235}U enrichment against the reactivity penalties of fuel depletion and Xenon-135 poisoning. By targeting $k_{eff} = 1$, the model successfully calculated that the B_4C control rods must be inserted at 19.9 % at the Start of Cycle (BOC) and gradually withdrawn to 5.3 % at the End of Cycle (EOC). This continuous optimization confirms that the initial fuel loading is mathematically sufficient to sustain the requested 2540 MW thermal output over the 1.5-year cycle, while still leaving a control margin for safe reactor shutdown.

An important parameter adjustment within the methodology was the implementation of a calibration multiplier (the scale factor) for the control rod volume. In a purely zero-dimensional core model, macroscopic cross sections are weighted exclusively by their relative volume fractions. That simplification fails to capture spatial flux depressions and the strong self-shielding effects characteristic of highly absorbing control blades. To bypass very complex calculations, the effective relative volume of the control rods was scaled down manually. This optimization calibrated the thermal utilization factor (f) to realistic bounds, preventing the control rods from artificially absorbing an exaggerated fraction of the thermal neutrons.

The moderator-to-fuel volume ratio (V_m/V_u) and the operational void fraction were also critical design parameters that determined the core's safety. By assuming $V_m/V_u = 3$ as we have done continuously in the course and analyzing the reactivity across a broad void spectrum, we verified that the reactor operates safely within an under-moderated regime at the 40 % void target. However, our results demonstrated that as the fuel depleted toward EOC, the negative void feedback weakened, with the reactivity peak shifting quite close to the operational setpoint. While the optimization of the con-

trol rod insertion successfully maintained criticality, decreasing the V_m/V_u ratio—would perhaps be advisable in a finalized design to broaden the negative void coefficient margin and enhance stability at EOC.

Finally, it is essential to contextualize the results by acknowledging the limitations of the fixed parameters. The model relies on static assumptions for the fast fission factor ($\epsilon = 1.06$), the total non-leakage probability ($P = 0.97$), and the macroscopic slowing-down power ($\xi\Sigma_s = 0.8\,\mathrm{cm}^{-1}$). Moreover, the calculation of the thermal fission factor (η) explicitly neglects the transmutation of Uranium-238 into fissile Plutonium-239. In a physical BWR, continuous plutonium breeding provides a substantial reactivity contribution toward the end of the operating cycle, meaning our model likely overestimates the required ^{235}U enrichment or underestimates the possible cycle length. While our model effectively demonstrates the fundamental physics of reactivity control and burnup, more advanced models would be required to fully optimize localized power distributions and dynamic thermohydraulic feedback.

5 Conclusion

We have modelled a BWR type reactor using the five-factor formula. Our results showcases that a stable critical reactor design is achieved, both at the start of a fuel cycle where $k_{eff} = 1$ is achieved with 3% enrichment and control rods 19.9% inserted, and at the end where $k_{eff} = 1$ is achieved with 1.65% enrichment and control rods 5.3% inserted. The stable criticality is the result of balancing the variables within the five-factor formula. While the fast fission factor ($\epsilon = 1.06$), resonance escape probability ($p \approx 0.82$), and non-leakage factor ($P = 0.97$) remain relatively constant throughout the cycle, the thermal utilization factor (f) increases from 0.64120 at BOC to 0.70486 at EOC. This strategic withdrawal of the control rods compensates for the thermal fission factor (η) dropping from 1.84851 to 1.68619 as the fuel depletes. The results also showcases a negative feedback coefficient for the reactor temperature, ensuring an inherently safe design.

6 Appendix

6.1 Python-code

Link to our Github is provided below:

https://github.com/wesslenlinus/Linus_Sigvard_Math_Problem_1_Nuclear